DATA ANALYTICS FOR BUSINESS

with Lizhen Xu, Ph.D.

Endogeneity

Learning Objectives

- Describe what is endogeneity
- Explain the causes of endogeneity
- Determine the direction of biases caused by endogeneity

What is Endogeneity?

- Revisit a regular linear regression model:

$$y_i = \beta_0 + \beta_1 \cdot x_{i1} + \cdots + \beta_k \cdot x_{ik} + u_i$$

  - Important assumption: $E(u_i|X) = 0$

- Endogeneity
  - One or more explanatory variables are *correlated* with the error term
  - Coefficient estimates are biased and inconsistent
  - Fail to draw correct *ceteris paribus* causal implications
  - <u>A common challenge facing most empirical analysis</u>!

- Sources of Endogeneity:
  - Omitted Variables
  - Measurement Errors
  - Simultaneous Equations
  - ……

Endogeneity: Intuitively Explained

- Endogeneity Example:

$$wage = \beta_0 + \beta_1 educ + \beta_2 abil + u$$

  - $abil$: an individual's innate ability
    - Typically <u>*unobservable*</u>
    - Would affect the dependent variable $wage$
    - May *correlate* with the explanatory variable $educ$ (which is essentially a *choice*)

$$wage = \beta_0 + \beta_1 educ + v$$

  - Error term contains omitted variables: $v = \beta_2 abil + u$
  - The error term v is now *correlated* with the explanatory variable $educ$
  - $educ$ is called ***endogenous*** (no longer ***exogenous***)

- Intuitive consequences of Endogeneity:
  - If $abil$ positively correlates with $educ$, and $abil$ positively affects $wage$
  - The effects of $educ$ on $wage$ will be overestimated

Endogeneity: Formally Explained

- Problems caused by Endogeneity:

$$wage = \beta_0 + \beta_1 educ + v$$

$$\hat{\beta}_1^{OLS} = \frac{Cov(x,y)}{Var(x)} = \frac{Cov(x, \beta_0 + \beta_1 x + v\,)}{Var(x)}$$
$$= \beta_1 + \frac{Cov(x,v)}{Var(x)}$$

- If $Cov(x,v) \neq 0$, $\hat{\beta}_1^{OLS}$ is <u>no longer an unbiased and consistent estimator</u> of β_1!

$$\hat{\beta}_1^{OLS} = \beta_1 + \frac{Cov(educ, \beta_2 abil + u)}{Var(educ)}$$
$$= \beta_1 + \beta_2 \frac{Cov(educ, abil)}{Var(educ)} = \beta_1 + \beta_2 \delta_1$$

- δ_1 is the slope of regressing $abil$ on $educ$

Endogeneity: Formally Explained (Cont'd)

- Endogeneity Issue in General Multiple Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_{k-1} x_{k-1} + (\beta_k x_k + u)$$

  - If x_k is unobservable and omitted, and we simply regress y on $x_1, \ldots, x_{k-1}$

$$\hat{\beta}_j^{OLS} = \beta_j + \beta_k \delta_j$$

  - δ_j is the slope coefficient of x_j from regressing x_k on $x_1, \ldots, x_{k-1}$
  - δ_j measures the *partial correlation* b/w x_k and x_j given all other variables

	$\delta_j > 0$	$\delta_j < 0$
$\beta_k > 0$	Positive bias	Negative bias
$\beta_k < 0$	Negative bias	Positive bias

Endogeneity is Common

- Unfortunately, endogeneity is way too common…
  - Student Grade ~ Class Participation (+ Self-Motivation)
  - Crime Rate ~ Police Force (+ District Characteristics)
  - Demand ~ Price (+ Demand Shocks)
  - Consumer Spending ~ Promotional Program Sign-up (+ Intrinsic Interest)
  - ……

- When is endogeneity a problem?
  - When we want to derive ***causal*** implications:
    - To measure the ***causal effect*** of a particular independent variable
    - When the focal *x* variable changes, holding everything else the same, how much will the outcome change?
  - Prediction is NOT affected:
    - As long as the error term distribution (and its potential correlation with *x* variables) remains the same for the new data used for prediction

Tackling Endogeneity

- Tackling Endogeneity
  - Proxy Control Variables
  - Panel Data and Fixed Effects (individual & time)
  - Matching Samples
  - Sensitivity Analysis
  - Instrumental Variable (IV)
  - Natural Exogenous Shocks
  - Randomized Controlled Experiments
  - ……

Summary

- Endogeneity overview
  - Examples
  - Causes
  - Symptoms
  - Treatments

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Instrumental Variables

Learning Objectives

- Explain the requirements for Instrumental Variables (IV)
- Apply IV estimation
- Apply Two-Stage Least Square (2SLS) estimation
- Explain the intuition behind 2SLS

Instrumental Variable

- Simple Regression Case:

$$y = \beta_0 + \beta_1 x + u$$

  - x is *endogenous*: $Cov(x, u) \neq 0$
  - If we can find a variable z, which is *exogenous*, $\boldsymbol{Cov(z, u) = 0}$

$$Cov(z, y - \beta_0 - \beta_1 x) = Cov(z, y) - \beta_1 Cov(z, x) = 0$$

$$\beta_1 = \frac{Cov(z, y)}{Cov(z, x)}$$

  - As long as $\boldsymbol{Cov(z, x) \neq 0}$, i.e., z is correlated with x:
  - We then have the **instrumental variable (IV) estimator** of β_1

$$\hat{\beta}_1^{IV} = \frac{\sum_{i=1}^{n}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{n}(z_i - \bar{z})(x_i - \bar{x})}$$

  - z is called the **instrumental variable**, or simply **instrument**
  - IV estimator $\hat{\beta}_1^{IV}$ is a *consistent* estimator of β_1
    - $\hat{\beta}_1^{IV}$ is different from $\hat{\beta}_1^{OLS}$, correcting the bias caused by endogeneity
    - If x is exogenous, then x can be its own instrument, and $\hat{\beta}_1^{IV} = \hat{\beta}_1^{OLS}$

IV Requirements

- <u>Requirements on instrumental variables</u>
- **Instrument Exogeneity**
  - The instrument needs to be *uncorrelated* with the error term (including any unobservable variables)
  - $\boldsymbol{Cov(z, u) = 0}$
- **Instrument Relevance**
  - The instrument needs to be *correlated* with the endogenous explanatory variable
  - $\boldsymbol{Cov(z, x) \neq 0}$
- Intuitively, IV should be:
  - Some external perturbation that has no effects on y *after* x *is controlled for*
  - But strongly correlates with x
- IV examples:
  - Return on education: for the endogenous variable *Years of Schooling*
  - Possible IV: *Proximity to a College*, *Quarter of Birth*

IV Estimator

- <u>Multiple Regression Case</u>:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \beta_{k+1} z_{k+1} + \cdots + \beta_{k+m} z_{k+m} + u$$

  - Instruments: $\boldsymbol{z_1, \ldots, z_k}$
  - The **IV estimator** $\hat{\beta}^{IV}$ solves the following system of $(k+m+1)$ equations

$$\begin{cases} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} - \hat{\beta}_{k+1} z_{i,k+1} - \cdots - \hat{\beta}_{k+m} z_{i,k+m}) = 0 \\ \sum_{i=1}^{n} \boldsymbol{z_{i1}} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} - \hat{\beta}_{k+1} z_{i,k+1} - \cdots - \hat{\beta}_{k+m} z_{i,k+m}) = 0 \\ \cdots \\ \sum_{i=1}^{n} \boldsymbol{z_{ik}} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} - \hat{\beta}_{k+1} z_{i,k+1} - \cdots - \hat{\beta}_{k+m} z_{i,k+m}) = 0 \\ \sum_{i=1}^{n} z_{i,k+1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} - \hat{\beta}_{k+1} z_{i,k+1} - \cdots - \hat{\beta}_{k+m} z_{i,k+m}) = 0 \\ \cdots \\ \sum_{i=1}^{n} z_{i,k+m} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} - \hat{\beta}_{k+1} z_{i,k+1} - \cdots - \hat{\beta}_{k+m} z_{i,k+m}) = 0 \end{cases}$$

Two Stage Least Square (2SLS)

- General Case of Multiple Regression:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 z_1 + \cdots + \beta_{k+1} z_k + u$$

  - x_1 is *endogenous*: $Cov(x_1, u) \neq 0$
  - $z_1, \ldots, z_k$ are *exogenous*: $Cov(z_j, u) = 0$
  - Additional *exogenous* instrumental variables $z_{k+1}, \ldots, z_{k+m}$ $(m \geq 1)$

Two Stage Least Square (2SLS) (Cont'd)

- <u>Two Stage Least Square (2SLS) Estimation</u>:
- **First Stage**
  - Regress the endogenous variable x_1 on ***all*** exogenous variables $z_1, \dots, z_{k+m}$
  - Obtained the fitted value $\hat{x}_1$

$$\hat{x}_1 = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \cdots + \hat{\pi}_k z_k + \hat{\pi}_{k+1} z_{k+1} + \cdots + \hat{\pi}_{k+m} z_{k+m}$$

- **Second Stage**
  - Replace x_1 with $\hat{x}_1$ in the original regression
  - Regress y on $\hat{x}_1, z_1, \dots, z_k$

$$\hat{y} = \hat{\beta}_0^{2SLS} + \hat{\beta}_1^{2SLS} \hat{x}_1 + \hat{\beta}_2^{2SLS} z_1 + \cdots + \hat{\beta}_{k+1}^{2SLS} z_k$$

- The **2SLS estimator** $\hat{\beta}^{2SLS}$ is a <u>*consistent*</u> estimator of β

Instrument Requirements

- 2SLS estimator is equivalent to IV estimator when
  - We use $z_1, \dots, z_k$ as instruments for themselves
  - We use $\hat{x}_1 = \hat{\pi}_0 + \hat{\pi}_1 z_1 + \cdots + \hat{\pi}_{k+m} z_{k+m}$ as the instrument for x_1
    - $\hat{x}_1$ is the linear combination of all z that is *most highly correlated* with x_1

- Requirements on the instruments:
  - The number of the additional instruments $z_{k+1}, \dots, z_{k+m}$ needs to be <u>at least the same</u> as the number of endogenous variables: $m \geq 1$
  - Having more instruments than endogenous variables is okay, *preferrable* in fact
  - **Exogeneity**: $Cov(z_j, u) = 0, \forall j$
  - **Relevance**: At least one of $\hat{\pi}_{k+1}, \dots, \hat{\pi}_{k+m}$ must be non-zero, to avoid *multicollinearity*
    - The higher the correlation, the better
  - <u>Avoid ***weak instruments***</u>, which are weakly (or not) correlated with the endogenous variable
    - It will increase the standard error, causing the estimates to vary wildly
    - May result in more biases than it aims to correct

Intuition behind 2SLS

- <u>Intuition behind 2SLS estimation</u>
  - First, "purge" x_1 of its correlation with u by decomposing it into two parts
    $$x_1 = \hat{x}_1 + v$$
    - $\hat{x}_1$, a linear combination of all instruments, is uncorrelated with the error term u
    - Residual v could be correlated with u, which is the source of endogeneity
  - Second, substitute $x_1 = \hat{x}_1 + v$ into the original regression
    $$y = \beta_0 + \beta_1 \hat{x}_1 + \beta_2 z_1 + \cdots + \beta_{k+1} z_k + (u + \beta_1 v)$$
    - The new error term is $e = u + \beta_1 v$
    - All explanatory variables now $\hat{x}_1, z_1, \ldots, z_k$ are uncorrelated with e (ensured by **instrument exogeneity**)
    - This ensures consistent estimation of β
  - To ensure $\hat{x}_1$ is not perfect collinear with $z_1, \ldots, z_k$
    - We need at least one additional instrument z_{k+1}
    - And its coefficient $\hat{\pi}_{k+1}$ (which measures the partial correlation) cannot be zero
    - This corresponds to the **instrument relevance** requirement

Summary

- Instrumental Variable
  - Exogeneity
  - Relevance
- IV estimation
- 2SLS estimation
  - Equivalence between 2SLS and IV estimation
  - Intuition